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August 6, 1975
SUPERSEDING
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FEDERAL SPECIFICATION

PLATE, SPOT TEST, White Porcelain

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers white porcelain spot test plates suitable for use in laboratory procedures.

1.2 Classification.

1.2.1 Sizes. The spot test plates shall be of the following sizes, as specified (see 6.1):

Size 1 - 3 depressions (cavities).
Size 2 - 12 depressions (cavities).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, for a part of this specification to the extent specified herein:

Federal Specifications

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Box, Setup.

Federal Standard

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standard:

C 242 - Ceramic Whitewares and Related Products, Standard Definitions of Terms Relating to.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Material. Spot plate shall be fabricated of high quality porcelain having an average thermal coefficient of linear expansion between 20° and 200° Centigrade (°C.) of 0.0000036 cm./cm./1°C. gradually increasing to 0.0000047 cm./cm./1°C. at 1000°C. Softening point shall be not less than 1400°C. and fusion point shall be not less than 1670°C. Porcelain shall be subjected to two firings to insure uniformity.

3.2 Style and design.

3.2.1 Size 1. Size 1 plates shall be fabricated of porcelain, shall be oblong and shall have three depressions in the upper surface. Except for the bottom surface, plates shall be completely glazed.

3.2.2 Size 2. Size 2 plates shall be fabricated of porcelain, shall be oblong and shall have twelve depressions in the upper surface. Except for the bottom surface, plates shall be completely glazed.

3.3 Dimensions. The dimensions and tolerances shall be as specified in Table I.

TABLE I Dimensions and tolerances (in millimeters)

Size	Length ± 2	Width ± 2	Thickness ± 0.5	Diameter of depression ± 1	Depth of depression ± 0.5
1	92	31	12	21	7
2	112	92	7	21	5

3.4 Performance.

3.4.1 Heat resistance. The glazed and unglazed surfaces shall not crack, craze or melt, when tested as described in 4.4.1.

3.4.2 Resistance to chemical attack. Resistance of glaze to chemical attack, as measured by loss of weight in boiling in a solution of sodium hydroxide, shall be not more than 0.144%, when tested as described in 4.4.2.

3.4.3 Glaze adhesion. Permanent adhesion of glaze to another glazed porcelain surface of the same material shall not occur below a temperature of 1150°C., when tested as described in 4.4.3.

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3.5 Identification marking. Plates shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.6 Workmanship. Workmanship shall be first class throughout. Plates shall be free of defects listed in Table IV and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the porcelain, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as specified in Table II. Unit of product for examination purposes shall be one plate.

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TABLE II For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III, except that the acceptance number shall be zero for all sample sizes. The unit of product shall be one plate.

TABLE III For tests

Characteristic	Requirement	Test procedure	Inspection level
Resistance to heat	3.4.1	4.4.1	S-2
Resistance to chemical attack	3.4.2	4.4.2	S-2
Glaze adhesion	3.4.3	4.4.3	S-2

4.3 Examination. The plates shall be examined to determine compliance with all requirements contained in this specification. The plates shall be examined for defects including, but not limited to, those listed in Table IV.

TABLE IV Classification of defects

Categories	Defects
Major	
101	Plate not free of crawling.
102	Plate not free of dry edging.
103	Plate not free of cracks.
105	Plate not free of blistering.
Minor	
201	Identification marking not present, complete, correct, legible or permanent.
202	Plate not free of pinholes.

NOTE: For definition of defect terms, refer to ASTM Designation C242.

4.3.1 Dimensional examination. The plates shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

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4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE V Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence or method of application.

4.4 Tests.

4.4.1 Glow. Test samples shall be heated to a bright red (approximately 1000°C.) in an electric muffle-type furnace. Samples shall be removed and allowed to cool by radiation to room temperature.

4.4.2 Resistance to chemical attack. Boil glazed portion of test samples in 10 percent solution of sodium hydroxide for 8 hours.

4.4.3 Glaze adhesion. Place one plate on top of another so that glazed surfaces are in contact with each other and heat to a temperature of 1150°C. for a period of 30 minutes and allow to cool by radiation to room temperature.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each plate shall be wrapped in kraft paper or other

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suitable wrapping material and packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except commercial colors are acceptable. Carton shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Intermediate package. Twenty four size 1 plates or twelve size 2 plates shall be packaged in a snug-fitting box constructed in accordance with PPP-B-676 or PPP-B-636, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. One hundred forty four size 1 plates or seventy two size 2 plates shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. One hundred forty four size 1 plates or seventy two size 2 plates shall be packed in an exterior container designed for a type load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Size required (see 1.2.1).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6640-0916

Civil Agency Coordinating Activities:
GSA-FSS
HEW-HSM, NIH
VA-DMS

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